THE FOURTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 1, 1941

Morning Session

1. Prove that the polynomial

$$(a - x)^6 - 3a(a - x)^5 + \tfrac{5}{2} a^2(a - x)^4 - \tfrac{1}{2} a^4(a - x)^2$$

takes only negative values for $0 < x < a$. (page 161)

2. Find the nth derivative with respect to x of

$$\int_0^x \left[1 + \frac{(x - t)}{1!} + \frac{(x - t)^2}{2!} + \cdots + \frac{(x - t)^{n-1}}{(n - 1)!}\right] e^{nt}\, dt.$$

(page 162)

3. A circle of *radius* a rolls in its plane along the x-axis. Show that the envelope of a diameter is a cycloid, coinciding with the cycloid traced out by a point on the circumference of a circle of *diameter* a, likewise rolling in its plane along the x-axis. (page 163)

4. Let the roots a, b, c of

$$f(x) \equiv x^3 + px^2 + qx + r = 0$$

be real, and let $a \leq b \leq c$. Prove that, if the interval (b, c) is divided into *six* equal parts, a root of $f'(x) = 0$ will lie in the *fourth* part counting from the end b. What will be the form of $f(x)$ if the root in question of $f'(x) = 0$ falls at either end of the *fourth* part? (page 165)

5. Show that the line which moves parallel to the plane $y = z$ and which intersects the two parabolas $y^2 = 2x$, $z = 0$ and $z^2 = 3x$, $y = 0$ sweeps out the surface

$$x = (y - z)\left(\frac{y}{2} - \frac{z}{3}\right).$$

(page 166)

6. If the x-coordinate $\bar{x}$ of the center of mass of the area lying between the

x-axis and the curve $y = f(x)$, $(f(x) > 0)$, and between the lines $x = 0$ and $x = a$ is given by

$$\bar{x} = g(a),$$

show that

$$f(x) = A \frac{g'(x)}{[x - g(x)]^2} e^{\int dx/(x - g(x))},$$

where A is a positive constant. (page 168)

7. Take either (i) or (ii).

(i) Prove that

$$\begin{vmatrix} 1 + a^2 - b^2 - c^2 & 2(ab + c) & 2(ca - b) \\ 2(ab - c) & 1 + b^2 - c^2 - a^2 & 2(bc + a) \\ 2(ca + b) & 2(bc - a) & 1 + c^2 - a^2 - b^2 \end{vmatrix}$$

$$= (1 + a^2 + b^2 + c^2)^3.$$ (page 169)

(ii) A semi-ellipsoid of revolution is formed by revolving about the x-axis the area lying within the first quadrant of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Show that this semi-ellipsoid will balance in stable equilibrium, with its vertex resting on a horizontal plane, when and only when

$$b\sqrt{8} \geq a\sqrt{5}.$$ (page 171)

Afternoon Session

8. A particle (x, y) moves so that its angular velocities about $(1, 0)$ and $(-1, 0)$ are equal in magnitude but opposite in sign. Prove that

$$y(x^2 + y^2 + 1)\, dx = x(x^2 + y^2 - 1)\, dy,$$

and verify that this is the differential equation of the family of rectangular hyperbolas passing through $(1, 0)$ and $(-1, 0)$ and having the origin as center. (page 173)

9. Evaluate the following limits:

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1^2}}+\frac{1}{\sqrt{n^2+2^2}}+\cdots+\frac{1}{\sqrt{n^2+n^2}}\right);$$

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\cdots+\frac{1}{\sqrt{n^2+n}}\right);$$

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\cdots+\frac{1}{\sqrt{n^2+n^2}}\right).$$

(page 175)

10. Find the differential equation satisfied by the product z of any two linearly independent integrals of the equation

$$y'' + y'P(x) + yQ(x) = 0.$$ (page 176)

11. Two perpendicular diameters of the ellipse

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

are given, and the two diameters conjugate to them are constructed. Show that the rectangular hyperbola passing through the ends of these conjugate diameters passes through the foci of the ellipse. (page 178)

12. A car is being driven so that its wheels, all of radius a feet, have an angular velocity of ω radians per second. A particle is thrown off from the tire of one of these wheels, where it is supposed that $a\omega^2 > g$. Neglecting the resistance of the air, show that the maximum height above the roadway which the particle can reach is

$$\frac{(a\omega + g\omega^{-1})^2}{2g}.$$ (page 179)

13. Assuming that $f(x)$ is continuous in the interval $(0, 1)$, prove that

$$\int_{x=0}^{x=1}\int_{y=x}^{y=1}\int_{z=x}^{z=y} f(x)f(y)f(z)\,dx\,dy\,dz = \frac{1}{3!}\left(\int_{t=0}^{t=1} f(t)\,dt\right)^3.$$

(page 180)

14. Take either (i) or (ii).

(i) Show that any solution $f(t)$ of the functional equation

$$f(x+y)f(x-y) = f(x)f(x) + f(y)f(y) - 1, \quad (x, y, \text{real}),$$

is such that

$$f''(t) = \pm m^2 f(t), \quad (m \text{ constant and } \geq 0),$$

assuming the existence and continuity of the second derivative. Deduce that $f(t)$ is one of the functions

$$\pm \cos mt, \qquad \pm \cosh mt.$$ (page 181)

(ii) With n constant values $a_1, a_2, \ldots, a_n$, supposed all different, let n constant values $b_1, b_2, \ldots, b_n$ be associated, and let a polynomial $P(x)$ be defined by the identity in x

$$\begin{vmatrix} 1 & x & x^2 & \cdots & x^{n-1} & P(x) \\ 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} & b_1 \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} & b_n \end{vmatrix} \equiv 0.$$

Given a polynomial $\phi(t)$, let a polynomial $Q(x)$ be defined by the identity in x obtained on replacing $P(x)$, $b_1, b_2, \ldots, b_n$ of the identity above by $Q(x)$, $\phi(b_1)$, $\phi(b_2)$, $\ldots$, $\phi(b_n)$. Prove that the remainder obtained on dividing $\phi(P(x))$ by $(x - a_1)(x - a_2) \cdots (x - a_n)$ is $Q(x)$. (page 182)